

GREG KNAPP

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CURRENT EMPLOYMENT

Colby College

July 2025–Present

Visiting Assistant Professor of Mathematics

- Current research topics: Thue equations, Diophantine inequalities, polynomial root separation.

FORMER EMPLOYMENT

University of Calgary

August 2023–July 2025

PIMS Postdoctoral Fellow

Supervisor: Dang-Khoa Nguyen

EDUCATION

University of Oregon

PhD, Mathematics

June 2023

Advisor: Shabnam Akhtari

Dissertation title: Polynomial Root Distribution and Its Impact on Solutions to Thue Equations

Case Western Reserve University

Bachelor of Science, Mathematics

May 2017

Bachelor of Arts, Philosophy

May 2017

Master of Science, Mathematics

August 2017

Advisor: Colin McLarty

Thesis title: Minkowski's Linear Forms Theorem in Elementary Function Arithmetic

RESEARCH EXPERIENCE

Research Papers

- Greg Knapp and Chi Hoi Yip. Upper bounds on polynomial root separation. *Bulletin of the Australian Mathematical Society*, 112(2):274–289, 2025.
- Seda Albayrak, Samprit Ghosh, Greg Knapp, and Khoa D. Nguyen. On certain polytopes associated to products of algebraic integer conjugates. *Proceedings of the American Mathematical Society*, 153:1917 – 1932, 2025.
- Greg Knapp. The number of solutions to the trinomial Thue equation. *Functiones et Approximatio Commentarii Mathematici*, 69(2):247 – 270, 2023.
- Greg Knapp. *Polynomial Root Distribution and Its Impact on Solutions to Thue Equations*. PhD thesis, University of Oregon, 2023.

- Greg Knapp. Minkowski's Linear Forms Theorem in Elementary Function Arithmetic. Master's thesis, Case Western Reserve University, 2017.

Works In Progress

- Relative Thue Equations (with Eva Goedhart and Sumin Leem).

Upcoming Talks

- AMS Special Session on Recent Trends in Analytic Number Theory II, 2026 Joint Mathematics Meetings, Washington, DC. January 2026.

Invited Talks

- Number Theory and Algebraic Geometry Seminar, Simon Fraser University. April 2025.
- Number Theory and Combinatorics Seminar, University of Lethbridge. January 2025.
- PIMS Postdoctoral Summit, University of Calgary, June 2024.
- Number Theory By Early Career Researchers, 2024 CMS Summer Meetings, University of Saskatoon, June 2024.
- Calgary Junior Math Contest Awards Ceremony, University of Calgary, June 2024
- PIMS CRG Seminar: L -functions in Analytic Number Theory, University of Lethbridge, April 2024.
- Alberta Number Theory Days, Banff International Research Center, March 2024.
- PIMS Postdoctoral Fellow Seminar, Online, October 2023.
- Oregon State University Number Theory Seminar, February 2023.
- University of British Columbia Number Theory Seminar, December 2022.
- Portland State University GDMNT, November 2022
- Oregon Number Theory Days, Oregon State University, February 2022.

Contributed Talks

- Special Session on Number Theory and Field Theory II, 2025 Joint Mathematics Meetings, Seattle, WA. January 2025
- Number Theory by Early Career Researchers, 2023 CMS Winter Meetings, Montréal, QC, December 2023.
- Oregon Number Theory Days, Oregon State University, February 2023.
- New York State Regional Graduate Mathematics Conference, Syracuse University, April 2022.
- Undergraduate Poster Session, Joint Mathematics Meetings, Atlanta, GA, January 2017.
- Young Mathematicians Conference, Ohio State University, August 2016.

Links to Recorded Talks

- "On certain polytopes associated to products of algebraic integer conjugates." Number Theory and Combinatorics Seminar. University of Lethbridge. Lethbridge, AB (January 2025). <https://www.mathtube.org/lecture/video/certain-polytopes-associated-products-algebraic-integer-conjugates>.

- “Bounds on the Number of Solutions to Thue Equations.” PIMS CRG Seminar: L -functions in Analytic Number Theory. University of Lethbridge. Lethbridge, AB (April 2024). <https://www.mathtube.org/lecture/video/bounds-number-solutions-thue-equations-0>.

TEACHING EXPERIENCE

Colby College

July 2025–Present

Visiting Assistant Professor of Mathematics

- Courses taught as instructor of record:

Course Title	Terms Taught
Single-Variable Calculus	F25
Linear Algebra	F25

University of Calgary

August 2023–July 2025

PIMS Postdoctoral Fellow and Sessional Instructor

- Courses taught as instructor of record:

Course Title	Terms Taught
Linear Methods I	F23, W24, F24
Introduction to Statistics I	Sp25

- Earned the following badges from the University of Calgary Taylor Institute for Teaching and Learning.

Course Title	Month Completed
Theories and Practices	December 2024
Scholarship of Teaching and Learning Foundations	June 2024
Developing Your Teaching Dossier	May 2024
Equity, Diversity, and Inclusion	April 2024

- Earned the Postdoctoral Certificate in University Teaching and Learning for completing those four badge courses.

University of Oregon

August 2017–June 2023

Graduate Employee

- Courses taught as instructor of record:

Course Title	Terms Taught
University Mathematics I	Su18, Su19
College Algebra	F17
Elementary Functions	Sp18
Calculus for Business and Social Sciences II	F18
Differential Calculus	W20, F20
Integral Calculus	Sp20, W21, F21
Sequences and Series	Sp21
Fundamentals of Number Theory	W22, Sp22

- Courses for which I have been a teaching assistant:

Course Title	Terms as TA
College Algebra	Sp19
Calculus for Business and Social Sciences I	W18, F19, Sp23, W23
Calculus for Business and Social Sciences II	W19

Case Western Reserve University

January 2015–May 2017

Peer Tutor

- Tutored peers for for about 8 hours per week in the following classes:
 - Calculus I, II, III
 - Differential Equations
 - Linear Algebra
 - Abstract Algebra
 - Intro to Logic
 - Math Logic and Model Theory
- Assisted in the creation of the spring 2016 training sessions.

Boonshoft Museum of Discovery

Summer 2013, 2014, 2015, 2017

Summer Educator

- Created and taught three week-long summer camps: a math/logic/computer science camp, a computer programming camp, and a creativity/teamwork focused camp.
- Taught public programs: designed and demonstrated science experiments to the museums' guests
- Served as a camp counselor for summer campers (ages 6–14): escorted campers to and from classes and positively interacted with the campers within their classes.

Notable Teaching Activities

- Winter 2024: Designed and implemented a series of exploratory exercises on math games like Nim and Chomp for a series of regular outreach events called Math Nites.
- Fall 2023: Designed and implemented weekly active learning sessions for my linear algebra class.

- Spring 2023: Co-designed and ran activities on math games like Set and Dots & Boxes for one session of the Eugene Girls' Science Adventure Days.
- Summer 2022: Co-ran a two week long problem solving session with incoming first-year PhD students at the University of Oregon to help them decide which classes to take during their first year.
- Spring 2022: Co-designed and ran activities on math magic for one session of the Eugene Girls' Science Adventure Days.
- Spring 2020: Designed and ran a fully asynchronous integral calculus class, with lecture videos you can find here: <https://www.youtube.com/@gregknapp1653>

MENTORING

PIMS First Year Interest Group

October 2023–May 2024

Diophantine Approximation

- Mentored a graduate-level reading project on Diophantine approximation.
- Met once every other week to discuss readings from Schmidt's *Diophantine Approximation* and Bilu and Bugeaud's *Démonstration du théorème de Baker-Feldman via les formes linéaires en deux logarithmes*.

University of Oregon Directed Reading Program

January 2023–April 2023

Continued Fractions

- Mentored an undergraduate reading project on the measure theory of continued fractions.
- Met once per week with my student to discuss the second half of A. Khichin's *Continued Fractions*.
- Advised the student's 20 minute presentation on Khinchin's Theorem.

University of Oregon Directed Reading Program

January 2020–April 2021

Formal Logic

- Mentored an undergraduate reading project in formal logic, ranging from the Gödel's Incompleteness Theorem to subsystems of Peano Arithmetic.
- Met roughly once per week with my student January–March 2020 and July 2020–December 2022.
- Student read selections from chapters 1, 2, and 4 of Goldstern and Judah's *The Incompleteness Phenomenon* and from chapter 1 of Hájek and Pudlák's *Metamathematics of First-Order Arithmetic*.
- Student gave a 20 minute presentation on the first-order system of Peano Arithmetic.

University of Oregon Department of Mathematics

August 2019–June 2023

First-Year Mentor

- Mentored first-year mathematics graduate students.
- Met formally once per quarter and informally more frequently with my mentees to help them during their first year of graduate school.

SERVICE

Outreach

- Letters to a Pre-Scientist *Pen pal* October 2024–June 2025
- University of Calgary Math Nite *Volunteer* September 2023–June 2025
- University of Oregon AWM Graduate Student Chapter *Co-chair of the K-12 Outreach Committee* August 2019–June 2023
- Eugene Girls' Science Adventure Days *Volunteer instructor* March 2022, 2023
- Eugene Math Festival *Volunteer activity leader* February 2020, 2022, 2023
- Boonshoft Museum of Discovery *Camp Counselor* June/July 2012 and July 2016

Leadership

- 2025 Alberta Number Theory Days *Co-organizer* September 2024–May 2025
- University of Calgary Number Nosh Seminar *Organizer* September 2023–June 2025
- University of Oregon Graduate Student Teaching Seminar *Co-organizer* December 2019–March 2023
- University of Oregon AMS Graduate Student Chapter *Co-founder and Secretary* July 2020–October 2022
- University of Oregon Student Number Theory Seminar *Organizer* March 2021–September 2022
- University of Oregon Math Department *Elected Student Representative to the Graduate Affairs Committee* August 2019–June 2021

Refereeing

- Research in Number Theory (1 paper refereed)
- Acta Arithmetica (1 paper refereed)
- Periodica Mathematica Hungarica (1 paper refereed)

HONORS AND AWARDS

PIMS/BIRS Team Up award recipient, November 24–December 7, 2024 (with Eva Goodhart and Sumin Leem).

University of Calgary Students' Union Teaching Excellence Award nominee for teaching excellence in my winter 2024 Linear Methods I class.

Anderson Graduate Teaching Award for teaching dedication and excellence as a graduate student (awarded in 2022 for my work in 2017–2022).

John Schoff Millis Award for the senior with the best academic record in the College of Arts and Sciences at CWRU.

Outstanding Poster Award for the top 10% of poster presentations in the Undergraduate Poster Session of the Joint Mathematics Meetings, 2017.

Case Alumni Association Junior-Senior Scholarship recipient.